NMC 2011 PRIMARY LEVEL PAPER 1 QUESTIONS

SECTION A : 5 marks each

Qn. 1. Put one addition sign between any two digits in 54321 so that the resulting sum is the largest.

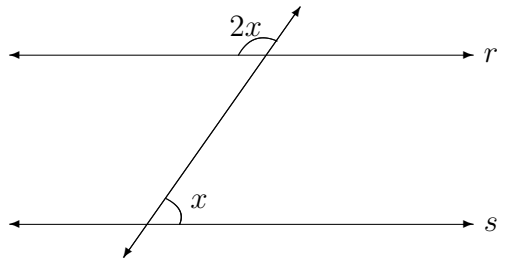
Qn. 2. Replace each blank circle with the correct digit in the following addition:

$$\begin{array}{cccc} & 3 & 9 & \bigcirc & 5 \\ & \bigcirc & 1 & 7 & \bigcirc \\ + & & 5 & 6 & 9 \\ \hline & 6 & 6 & 6 & 6 \\ \hline\hline \end{array}$$

Qn. 3. Given that Anna is older than Ben, Ben is older than Isaac, Isaac is younger than Anna and Jose is older than Anna. List the people from the oldest to the youngest.

Qn. 4. Four boys work together painting houses. For each house they paint, they get Shs. 512, 000. If the boys work for 4 months and their expenses are Shs. 304, 000 per month, how many houses must they paint for each one of them to have Shs. 2, 000, 000 at the end of the 4 months?

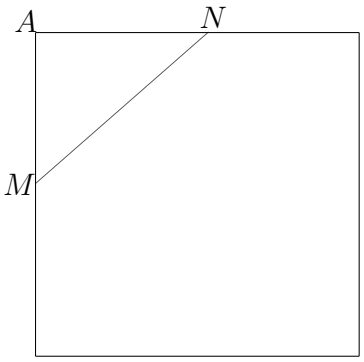
Qn. 5.



Lines r and s are parallel, x equals _____

Qn. 6. A man can walk 4Km in 1 hour and his wife can walk 5Km in 1 hour. They begin walking at the same time from opposite ends of a 12Km road. They have agreed to meet at the half way point. For how many minutes will one wait for the other to arrive at the agreed meeting point?

Qn. 7. M and N are the midpoints of the sides of a square. What is the ratio of the area of triangle AMN to the area of the complete square?



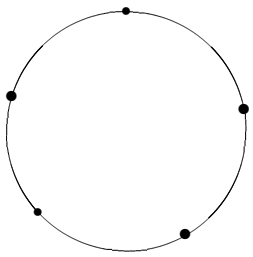
Qn. 8. Farida made a list of all the whole numbers between 1 and 100. How many times did she write the number 2?

Qn. 9. The average mark for 40 pupils is 47. How many free marks must the teacher add to each pupil to raise the average mark to 50?

Qn. 10. The length of a side of square B is four times the length of a side of square A. How many times greater is the area of square B than that of square A?

SECTION B : 10 **marks each**

Qn. 11. How many triangles can you draw using any three of the five points on the circle as vertices?



Qn. 12. The number of times a dog barks depends on the number of people passing by her home. Given that the following barks correspond to the number of people that passed her home:

People	Barks
6	4
7	7
8	10
9	13
10	16
11	19
12	22

(a) How many people have passed when a dog barks 31 times?

(b) How many barks for 5 people?

Qn. 13. Uganda Telecom(UTL) unveiled a new product that has enabled its customers to reduce on telephone call expenses. First of all, it has cut its call rates to Ug Shs 3 per second across all networks. In addition, it unveiled what is called **ENDOBO Profile.** To subscribe to ENDOBO, you pay Ug Shs 1000 and this gives you an opportunity to call all UTL mobile and land line telephones for free till midnight from the time you subscribe. For all other networks, the standard Ug Shs 3 per second still stands. On a given day, after subscribing for ENDOBO, you made 25 calls to UTL land line and you spoke for 2 minutes each. You also called 12 UTL mobile lines and spoke for $2\frac{1}{2}$ minutes each. In addition, you talked to 18 friends who were on ZAIN and for each friend, you used $2\frac{1}{2}$ minutes. How much money did you spend on telephone on this day? If you had forgotten to subscribe to ENDOBO, how much money would you have used for UTL calls?

Qn. 14. Below are charges displayed at Nelson Mandela National Stadium Nambole for three games:

	ADULTS	CHILDREN
SOFTBALL	10,000	5,000
BASKETBALL	15,000	7,500
SOCCER	25,000	12,500

(a) Mr. and Mrs. Mukasa and their 3 children paid Shs. 35,000 for tickets to watch the same game. Which game did they pay for? Give reasons for your answer.

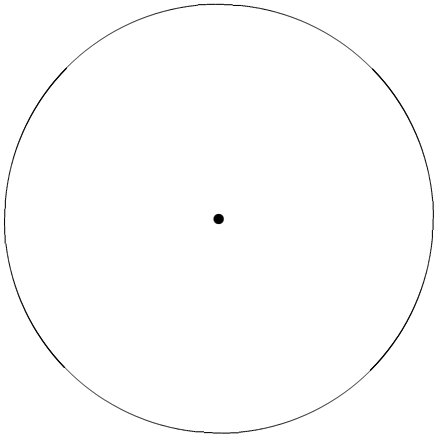
(b) Mr. and Mrs. Mukasa together with their children went to watch the basketball game and paid for it with two Shs. 50,000 notes. How much change did they get?

Qn. 15. Four friends buy a circular pizza that costs Shs. 40,000. Each person contributes the following amount of money:

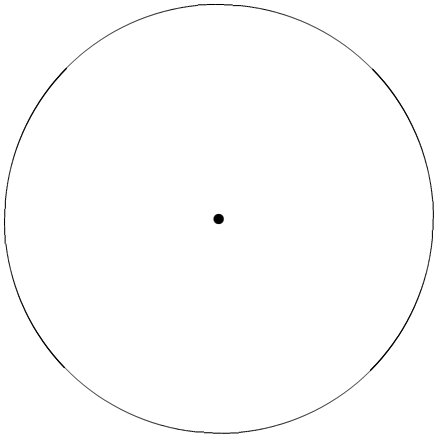
Joan	10,000
Peter	16,000
Mercy	6,000
Paul	8,000

Each person will eat a piece proportional to the amount of money they paid.

(a) The circle below represents the pizza. Divide the pizza among the four friends by showing the fraction (piece) each will get (show your working).



(b) Peter only eats half of his share he received from part (a) above and gives half of what he has to Mercy. Represent the new share of the pizza on the circle below:



NMC 2011 PRIMARY LEVEL PAPER 1 SOLUTIONS

SECTION A : 5 marks each

NMC 2011 PRIMARY LEVEL PAPER 2 SOLUTIONS

Qn1. $-9, -2, \frac{1}{3}, \frac{1}{2}, \frac{3}{4}, 0.8, 1.$

Qn2. (a) 9001 grams.

(b) 4 metres.

(c) $\frac{3}{2}$ metres.

Qn3. We are given that $\frac{3}{5} = \frac{3}{10}$. If we multiply both sides by $\frac{5}{3}$ we get $x = \frac{3}{10} \times \frac{5}{3} = \frac{5}{10} = \frac{1}{2}$.

Qn4.

a	3	b	3	4
3	4	7	c	d
8	e	3	3	7

$4 + d = 7$, so $d = 7 - 4 = 3$. $3 + c = c$, so $c = 3 - 3 = 0$. $b + 7 = 13$, so $b = 13 - 7 = 6$. To get e note that I carried 1, so $e = 4 + 4 = 8$. Lastly, $a + 3 = 8$, so $a = 8 - 3 = 5$.

Qn5. The two kids together ate $2 \times 2 = 4$ pieces. The number of pieces remaining for the adults are $32 - 4 = 28$. Each adult ate 4 pieces, so there were $\frac{28}{4} = 7$ adults.

Qn6. $\angle ACM + \angle CAM + \angle AMC = 180$. Therefore, $\angle AMC = 180 - 56 - 78 = 46^o$. $\angle DMB = \angle AMC$ because they are alternate angles.

Since the all angles $\triangle DBM$ add up to 180° we have $\angle DMB + 62 + \angle DMB = 180$. Or $\angle DBM = 180 - 62 - \angle DMB = 180 - 62 - 46 = 72^o$.,

Qn7. $\frac{124}{8} = 15.5$ Since we cannot have 0.5 of a van, the smallest number of vans is 16.

Qn8. Let x be the number of rabbits. This means that there are $x + 4$ chicken. Each rabbit has four legs meaning that the x rabbits have $4x$ legs. Each chicken has 2 legs meaning that all chicken have $2(x+)4$ legs. The total number of legs being $4x + 2(x + 4) = 4x + 2x + 8 = 6x + 8$. We solve $6x + 8 = 44$ or $6x = 44 - 8 = 36$ or $x = \frac{36}{6} = 6$. Therefore, there are 6 rabbits and $6 + 4 = 10$ chickens.

Qn9. Each square has area $A = \text{side}^2 = (4cm)^2 = 16cm^2$. The area of the shaded shape is equal to $5 \times A = 5 \times 16cm^2 = 80cm^2$.

Qn10.

$$\begin{aligned}
\text{Area of the remaining Octagon} &= \text{Area of the square} - \text{Area of the four triangles} \\
&= \text{side}^2 - 4 \times \frac{1}{2}bh \\
&= (20cm)^2 - 4 \times \frac{1}{2} \times 4cm \times 4cm \\
&= 400cm^2 - 2 \times 4cm \times 4cm = 400cm^2 - 32cm^2 = 368cm^2.
\end{aligned}$$

SECTION B

Qn11. There are 8 triangles.

Qn12. $50\% = \frac{50}{100} = \frac{1}{2}$. So $\frac{1}{2}$ of the pupils are younger than 10 years, $\frac{1}{20}$ are 10 years old and $\frac{1}{10}$ are older than 10 years but younger than 12 years. Let x be the total number of pupils. This means that pupils who are younger than 12 years are $\frac{1}{2}x + \frac{1}{20}x + \frac{1}{10}x = \frac{10+2+1}{20}x = \frac{13}{20}x$. Those who are 12 or older than 12 years are $x - \frac{13}{20}x = \frac{7}{20}x$. We are given that those who are 12 years and above are 70, so we solve $\frac{7}{20}x = 70$, meaning that $x = 70 \times \frac{20}{7} = 200$. Therefore, the number of pupils who are 10 years old is $\frac{1}{10}x = \frac{1}{10} \times 200 = 20$.

Qn13. (a) We use the formula Speed $= \frac{\text{Distance Covered}}{\text{Time taken to cover the distance}}$ or

Time taken $= \frac{\text{Distance Covered}}{\text{Speed}}$. The time taken to cover half of the distance is $\frac{500}{100} = 5$

hours. The time taken to cover the other half is $\frac{500}{80} = 6$ hours and 25 minutes. Total time taken including the resting time is $5hrs + 6hrs + 25mins + 35mins = 12hrs$.

(b) If she driven non-stop at $100km$ per hour, she would have taken $\frac{1000}{100} = 10$ hours.

Qn14. (a) It is open 9 hours for 5 days, which in total comes to $9 \times 5 = 45$ hours. On Saturday, it is open for 5 hours. So, if there are no public holidays, the library is open for $45 + 5 = 50$ hours in a week.

(b) April has 30 days, 5 of which were Sundays, 2 were public holidays(Good Friday and Easter Monday) and 4 were Saturdays. So, the library was open for only 19 weekdays, for a total of $19 \times 9 = 171$ hours. For all the 4 Saturdays, the library was opened for $4 \times 5 = 20$ hours. So, in April, the library was opened for $171 + 20 = 191$ hours.

Qn15. (a) There were $51 + 61 + 49 + 63 = 224$ pupils in the $P7$ class.

(b) Those who did not choose Mathematics as their favorite subject are $51 + 49 + 63 = 163$. The percentage is $\frac{163}{224} \times 100\% = 72.8\%$.

(c) Each stream has $\frac{224}{4} = 56$ Pupils.